# groupwork9

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```
credit$Student <- ifelse(credit$Student == "Yes", 1, 0)
credit$Own <- ifelse(credit$Own == "Yes", 1, 0)
credit$Married <- ifelse(credit$Married == "Yes", 1, 0)
credit$Region <- as.numeric(factor(credit$Region, levels = c("East", "South", "West")))
```

# Question 1

Income and Rating were selected for the best 2 feature model.

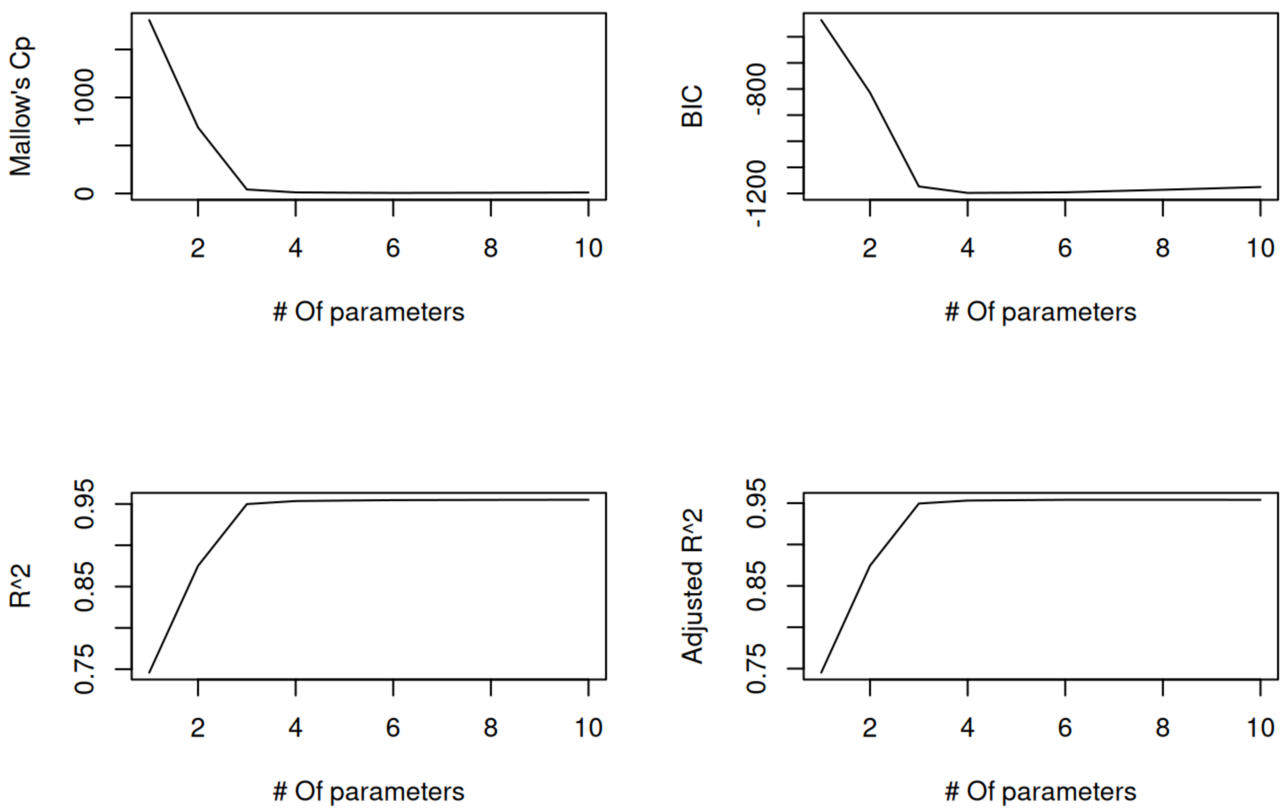
```
formula = as.formula(paste("Balance ~ ", paste(names(credit)[0:10], collapse= "+")))
credit_regfit = regsubsets(formula, data=credit, nvmax = NULL)
summary(credit_regfit)
```

```
## Subset selection object
## Call: regsubsets.formula(formula, data = credit, nvmax = NULL)
## 10 Variables  (and intercept)
##           Forced in Forced out
## Income        FALSE      FALSE
## Limit         FALSE      FALSE
## Rating        FALSE      FALSE
## Cards         FALSE      FALSE
## Age           FALSE      FALSE
## Education     FALSE      FALSE
## Own           FALSE      FALSE
## Student       FALSE      FALSE
## Married       FALSE      FALSE
## Region        FALSE      FALSE
## 1 subsets of each size up to 10
## Selection Algorithm: exhaustive
##           Income Limit Rating Cards Age Education Own Student Married Region
## 1  ( 1 )  " "    " "   "*"    " "   " " " "       " " " "     " "     " "
## 2  ( 1 )  "*"    " "   "*"    " "   " " " "       " " " "     " "     " "
## 3  ( 1 )  "*"    " "   "*"    " "   " " " "       " " "*"     " "     " "
## 4  ( 1 )  "*"    "*"   " "    "*"   " " " "       " " "*"     " "     " "
## 5  ( 1 )  "*"    "*"   "*"    "*"   " " " "       " " "*"     " "     " "
## 6  ( 1 )  "*"    "*"   "*"    "*"   "*" " "       " " "*"     " "     " "
## 7  ( 1 )  "*"    "*"   "*"    "*"   "*" " "       " " "*"     " "     "*"
## 8  ( 1 )  "*"    "*"   "*"    "*"   "*" " "       "*" "*"     " "     "*"
## 9  ( 1 )  "*"    "*"   "*"    "*"   "*" " "       "*" "*"     "*"     "*"
## 10  ( 1 ) "*"    "*"   "*"    "*"   "*" "*"       "*" "*"     "*"     "*"
```

# Question 2, 3

I think BIC and Cp are both choosing 6 variable model which are: Income, Limit, Rating, Cards, Age, Student

```
par(mfrow=c(2,2))
sum.fit.full=summary(credit_regfit)
plot(1:10,sum.fit.full$cp,type="l",col="black",xlab="# Of parameters",ylab="Mallow's Cp")
plot(1:10,sum.fit.full$bic,type="l",col="black",xlab="# Of parameters",ylab="BIC")
plot(1:10,sum.fit.full$rsq,type="l",col="black",xlab="# Of parameters",ylab="R^2")
plot(1:10,sum.fit.full$adjr2,type="l",col="black",xlab="# Of parameters",ylab="Adjusted R^2")
```




```
print(sum.fit.full$cp)
```

```
## [1] 1805.800985  687.895393   42.217059   12.152112    9.122228    6.554143
## [7]    7.410780    8.239154    9.482497   11.000000
```

```
print(sum.fit.full$bic)
```

```
## [1]  -535.9468  -814.1798 -1173.3585 -1198.0527 -1197.0957 -1195.7321
## [7] -1190.9074 -1186.1150 -1180.8999 -1175.4043
```

```
coef(credit_regfit,id=6)
```

```
##  (Intercept)       Income        Limit       Rating        Cards          Age
## -493.7341870   -7.7950824    0.1936914    1.0911874   18.2118976   -0.6240560
##      Student
##  425.6099369
```

# Question 4

looCV is also choosing 6 variable model with Income, Limit, Rating, Cards, Age, Student

```
#loocv function
looCV=function(fla,df){
  yhat=numeric(nrow(df))

  for (j in 1:nrow(df)){
    train=df[-j,]
    train_lm=lm(fla,data=train)
    yhat[j]=predict(train_lm,newdata=df[j,])
  }

  y=df[[all.vars(fla)[1]]]
  sqrt(mean((y-yhat)^2))
}


rmse=numeric(10)
for (i in 1:10){
formulas = as.formula(paste("Balance ~ ",
                            paste(names(coef(credit_regfit,id=i))[2:(i+1)],
                                  collapse= "+")))
rmse[i]=looCV(formulas, credit)
}
plot(1:10, rmse, color="black", xlab="Number of Terms")
```

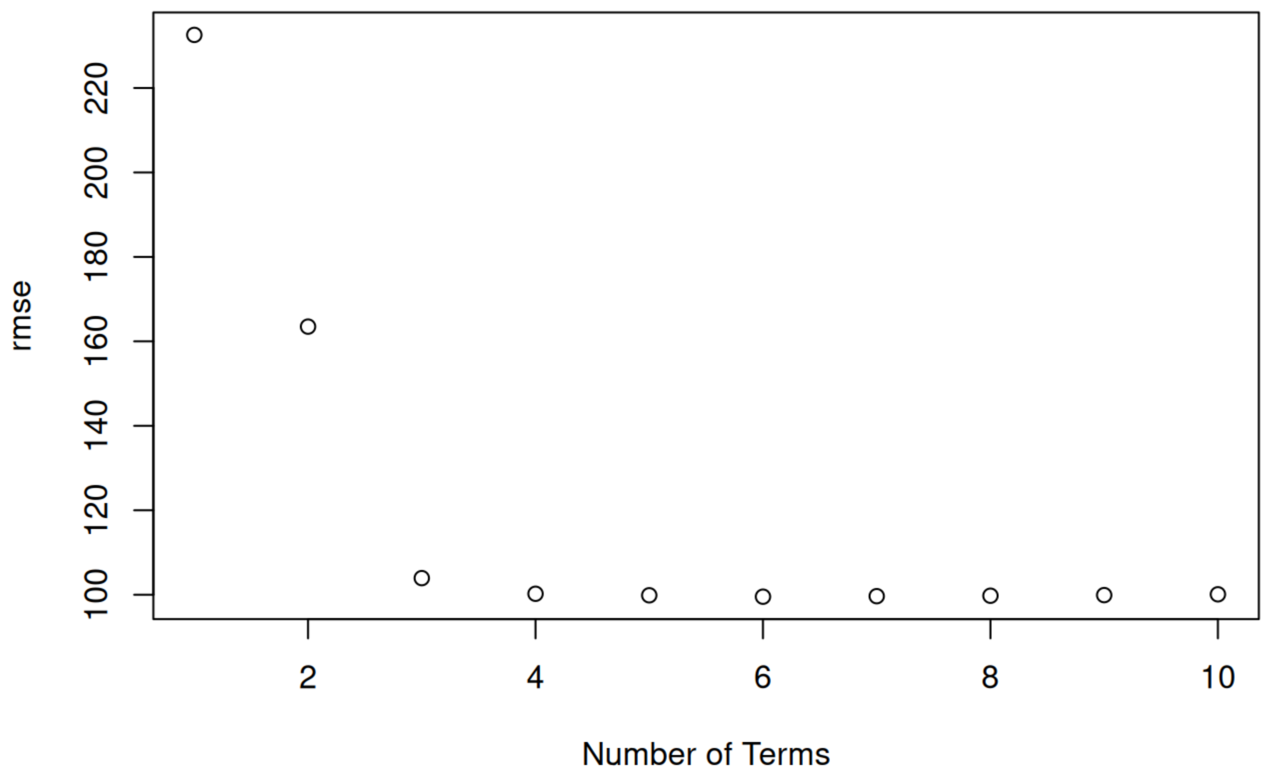
```
## Warning in plot.window(...): "color" is not a graphical parameter
```

```
## Warning in plot.xy(xy, type, ...): "color" is not a graphical parameter
```

```
## Warning in axis(side = side, at = at, labels = labels, ...): "color" is not a
## graphical parameter
## Warning in axis(side = side, at = at, labels = labels, ...): "color" is not a
## graphical parameter
```

```
## Warning in box(...): "color" is not a graphical parameter
```

```
## Warning in title(...): "color" is not a graphical parameter
```




```
print(rmse)
```

```
##  [1] 232.56406 163.49321 103.93831 100.23352  99.87055  99.54291  99.66226
##  [8]  99.77049  99.91537 100.10393
```